

JLP Data Science Technical Task

Problem definition

Customers receiving relevant and personalised communications from the John Lewis Partnership is key.

To ensure we are providing relevant content to customers, we need to know which products and product categories a customer is likely to purchase within a specific time frame. This way we can email customers about content, products or offers in areas they are interested in.

For this task, we want you to build a model to predict the likelihood of a customer purchasing within the **'Home'** category.

Instructions

1. Please use Python and any Python libraries of your choosing to complete this task and include an explanation of your approach. We would recommend using a Jupyter Notebook so that you can embed your explanations alongside your code. Please submit your solution as a rendered PDF if you do this.
2. Perform any exploratory data analysis that is required to understand the data and the problem space.
3. Create a model to predict the likelihood of a customer shopping in the 'Home' category.
4. Evaluate this model formally and describe how you might improve it.
5. Please do not spend more than 3 hours on this task. Please include a description of further ideas and/or areas that would warrant further investigation if you had more time to work on this.
6. Please submit your solution to this problem no later than 7 calendar days prior to your interview date.
7. Please come prepared to give a 10 minute presentation on this task in the interview, in whatever format you choose, which will be followed by 10 minutes of questions.

Data Description and Schema

Description:

Note: There may be issues in the data which need addressing or cleaning before modelling. The data is collected as a snapshot taken at a specific point in time. The target variable is binary and indicates if a customer has made a purchase within the **'Home'** category within

28 days **after** the snapshot date. The features reflect customer behaviour and characteristics **before** the snapshot date.

target.csv

Column Name	Type	Description
customer_id	String	Unique customer identifier
target	Binary	Binary target column: 1 indicates customers who have purchased within the 'Home' category, 0 indicates customers who have not made a purchase in the category.

demographic.csv

Column Name	Type	Description
customer_id	String	Unique customer identifier
affluence	Ordinal	Affluence at a household level from 1 (low) to 8 (high).
hh_size	Ordinal	Number of people in the household: 1: One person 2: Two people 3: Three people 4: Four people and above
customer_segment	Ordinal	Customer segment from 1 (low) to 7 (high) opportunity.

browsing.csv

Column Name	Type	Description
customer_id	String	Unique customer identifier
page_view	String	Path of the page viewed on the John Lewis website

customer_sales.csv

Column Name	Type	Description
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customer_id	String	Unique customer identifier
spend_cat_home	Numeric	Total spend in the 'Home' category before the snapshot date.
spend_cat_garden	Numeric	As above for the 'Garden' category.
spend_cat_sports	Numeric	As above for the 'Sports' category.
spend_cat_clothing	Numeric	As above for the 'Clothing' category.
spend_cat_gifts	Numeric	As above for the 'Gifts' category.
spend_cat_beauty	Numeric	As above for the 'Beauty' category.
spend_cat_shoes	Numeric	As above for the 'Shoes' category.
spend_cat_garden_accessories	Numeric	As above for the 'Garden Accessories' category.
transactions_cat_home	Numeric	Total number of transactions in the 'Home' category before the snapshot date.
transactions_cat_garden	Numeric	As above for the 'Garden' category.
transactions_cat_sports	Numeric	As above for the 'Sports' category.
transactions_cat_clothing	Numeric	As above for the 'Clothing' category.
transactions_cat_gifts	Numeric	As above for the 'Gifts' category.
transactions_cat_beauty	Numeric	As above for the 'Beauty' category.
transactions_cat_shoes	Numeric	As above for the 'Shoes' category.
transactions_cat_garden_accessories	Numeric	As above for the 'Garden Accessories' category.
last_purchase_date_home	String (date)	The date of the latest purchase in the 'Home' category before the snapshot date.
last_purchase_date_garden	String (date)	As above for the 'Garden' category.
last_purchase_date_sports	String (date)	As above for the 'Sports' category.
last_purchase_date_clothing	String (date)	As above for the 'Clothing' category.
last_purchase_date_gifts	String (date)	As above for the 'Gifts' category.
last_purchase_date_beau	String (date)	As above for the 'Beauty' category.

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last_purchase_date_shoes	String (date)	As above for the 'Shoes' category.
last_purchase_date_garden_accessories	String (date)	As above for the 'Garden Accessories' category.
total_spend	Numeric	The total spend within all the categories.
total_transactions	Numeric	The total number of transactions within all the categories.
last_purchase_date_all	String (date)	The date of the latest purchase across all the categories before the snapshot date.

additional.csv

Column Name	Type	Description
customer_id	String	Unique customer identifier
loyalty_customer	Binary	Binary column indicating if the customer is on the loyalty schema. 1 indicates they are, 0 indicates they are not.
closest_store	Categorical	The store they are closest to based on their home address.
distance_from_store	Numeric	Distance to the closest John Lewis store in miles.
jl_tenure	Numeric	Number of days they have been a John Lewis customer.
wtr_spend	Numeric	Total spend in Waitrose.
wtr_tenure	Numeric	Number of days they have been a Waitrose customer.